

Lecture - 18

09 March 2026

For Mid-Sem Exam:

* Syllabus: Lectures 1 - 17

* Venue: Chem Engg. LH1

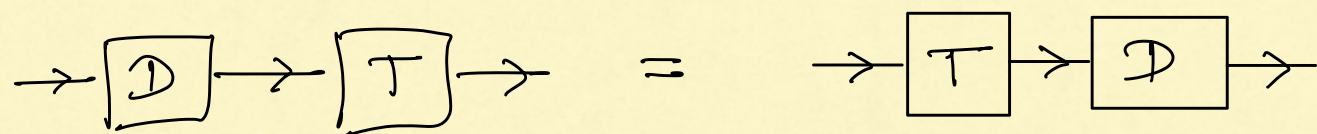
* Time: 6.30 pm (10 Mar)

* No calculators, electronic devices.

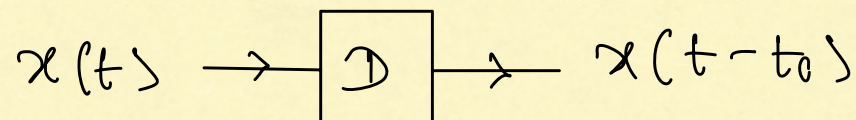
I. Time Invariance

Defn: A system is time invariant if it commutes with the time shift operator.

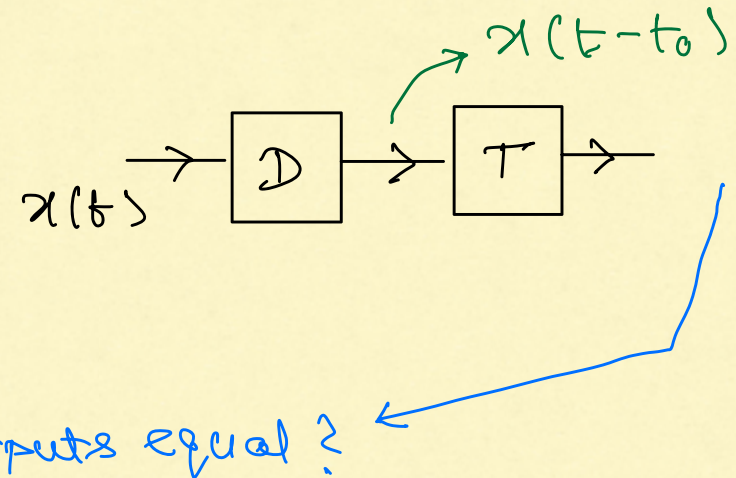
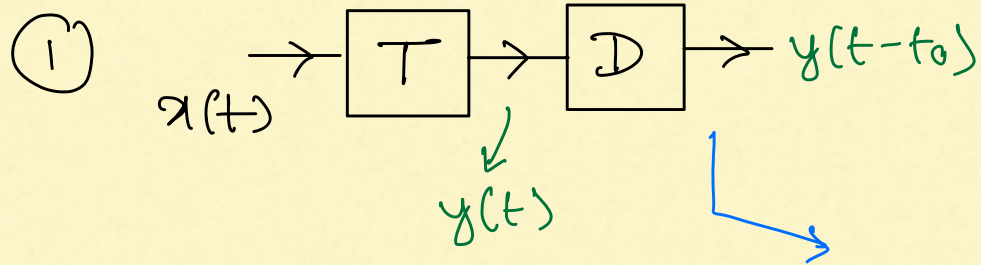
Alternatively, any time shift on the input must be effected as equal time shift on the output.



Here $\mathcal{D}$ denotes the time shift operator



Checking for time invariance



Are the outputs equal?

② Equivalently:

Assume: $x(t) \rightarrow [T] \rightarrow y(t)$

Check: $x(t-t_0) \rightarrow [T] \rightarrow$ Is output $= y(t-t_0)$?

Outputs must be equal for every x and t_0

Examples (Examples 1 and 2 in Lec 17)

③ T is a time shift operator

$$x(t) \rightarrow \boxed{T} \rightarrow x(t - t_1) \quad , \quad t_1 \text{ is a constant.}$$

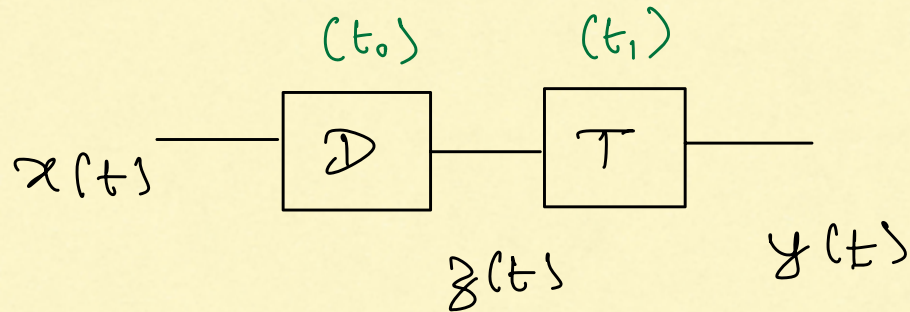
check for time invariance.

Examples (Examples 1 and 2 in Lec 17)

③ T is a time shift operator

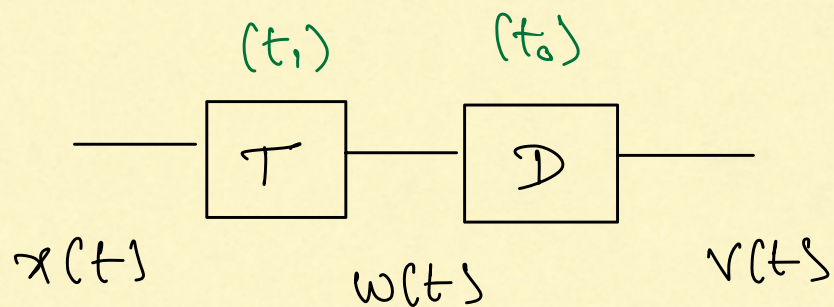
$$x(t) \rightarrow \boxed{T} \rightarrow x(t - t_1) \quad , \quad t_1 \text{ is a constant.}$$

Check for time invariance.



$$z(t) =$$

$$y(t) =$$



$$w(t) =$$

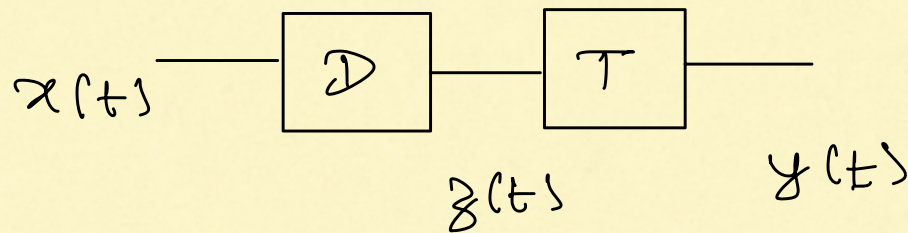
$$v(t) =$$

Examples (Examples 1 and 2 in Lec 17)

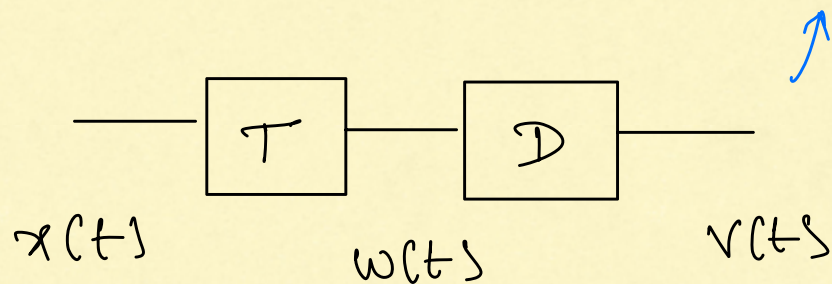
③ T is a time shift operator

$$x(t) \rightarrow \boxed{T} \rightarrow x(t-t_1), \quad t_1 \text{ is a constant.}$$

Check for time invariance.



↓
same output



$$z(t) = x(t-t_0)$$

$$\begin{aligned} y(t) &= z(t-t_1) \\ &= x(t-t_0-t_1) \end{aligned}$$

$$w(t) = x(t-t_1)$$

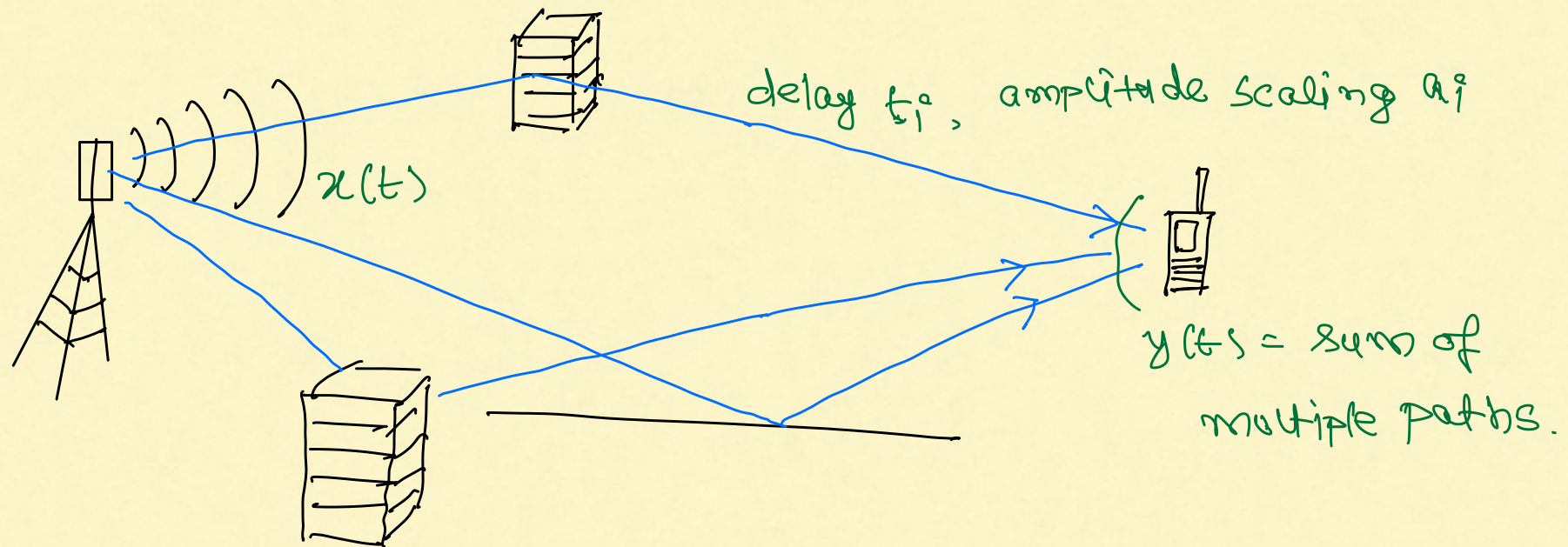
$$\begin{aligned} v(t) &= w(t-t_0) \\ &= x(t-t_0-t_1) \end{aligned}$$

$\therefore T$ is time invariant.

④

$$x(t) \rightarrow \boxed{T} \rightarrow y(t) = a_1 x(t-t_1) + a_2 x(t-t_2) + \dots + a_k x(t-t_k)$$

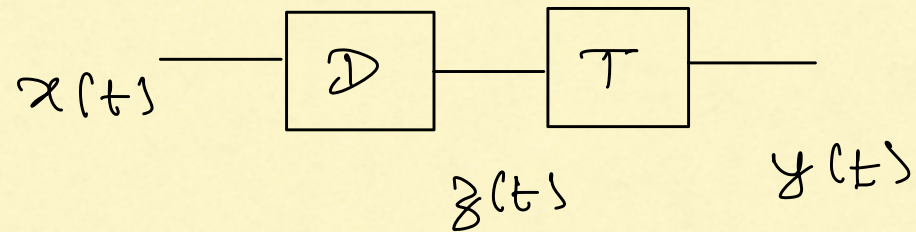
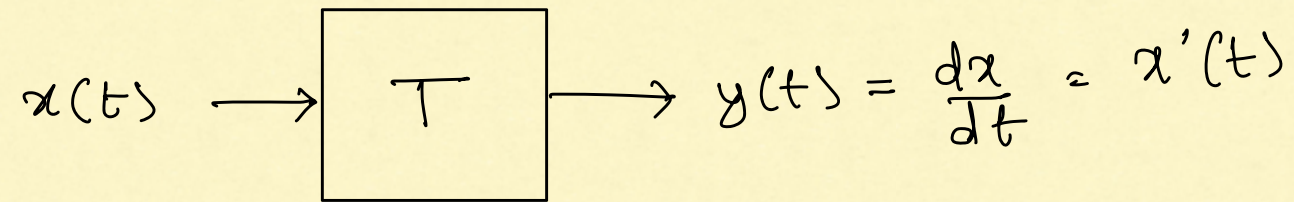
Here, $a_1, \dots, a_k$ and $t_1, \dots, t_k$ are constants.



Easy to check that this system is time-invariant.

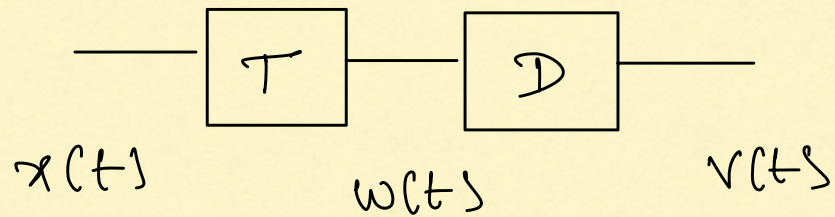
(Proof is similar to Example 3).

5a



$$z(t) =$$

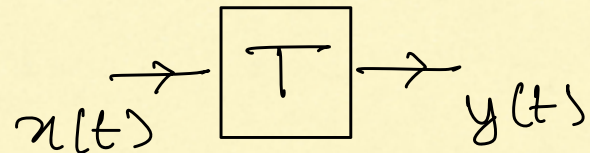
$$y(t) =$$



$$\omega(t) =$$

$$v(t) =$$

5b



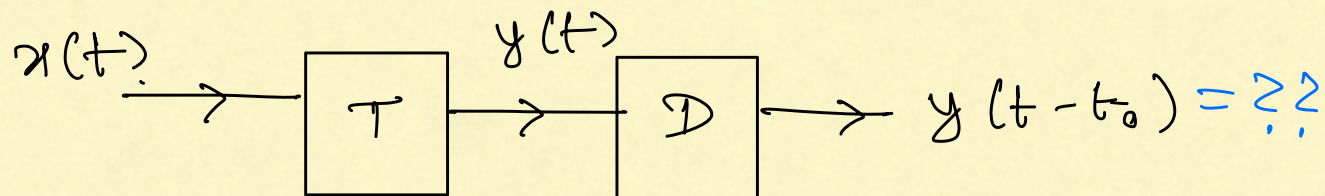
$$a_0 y(t) + a_1 y'(t) + \dots + a_n y^{(n)}(t)$$

$$= b_0 x(t) + b_1 x'(t) + \dots + b_m x^{(m)}(t)$$

Here, $a_0, \dots, a_n$ and $b_0, \dots, b_m$ are constants.

This is a system characterized by: linear constant coefficient differential equation.

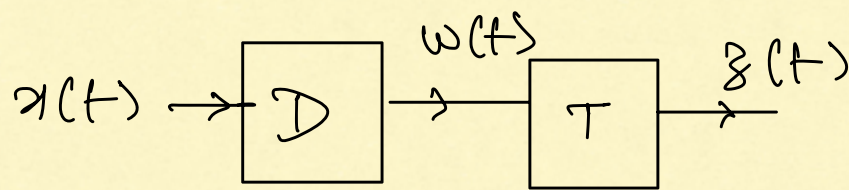
(Ref: Section 4-7 of OWN)



$$\begin{aligned}
 & a_0 y(t) + a_1 y'(t) + \dots + a_N y^{(N)}(t) \\
 & = b_0 x(t) + b_1 x'(t) + \dots + b_m x^{(m)}(t)
 \end{aligned}$$

This immediately implies that :

$$\begin{aligned}
 & a_0 y(t-t_0) + a_1 y'(t-t_0) + \dots + a_N y^{(N)}(t-t_0) \\
 & = b_0 x(t-t_0) + b_1 x'(t-t_0) + \dots + b_m x^{(m)}(t-t_0). \quad \text{--- (1)}
 \end{aligned}$$



Here, $w(t) = x(t-t_0)$

$$\Rightarrow w^{(k)}(t) = x^{(k)}(t-t_0)$$

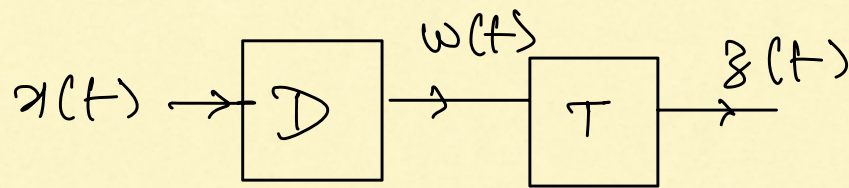
$$\begin{aligned}
 & a_0 z(t) + a_1 z'(t) + \dots + a_N z^{(N)}(t) \\
 & = b_0 w(t) + b_1 w'(t) + \dots + b_m w^{(m)}(t).
 \end{aligned}$$

$$= b_0 x(t-t_0) + b_1 x'(t-t_0) + \dots + b_m x^{(m)}(t-t_0) \quad \text{--- (2)}$$

$$\begin{aligned}
 & a_0 y(t) + a_1 y'(t) + \dots + a_N y^{(N)}(t) \\
 &= b_0 x(t) + b_1 x'(t) + \dots + b_M x^{(M)}(t)
 \end{aligned}$$

This immediately implies that :

$$\begin{aligned}
 & a_0 y(t-t_0) + a_1 y'(t-t_0) + \dots + a_N y^{(N)}(t-t_0) \\
 &= b_0 x(t-t_0) + b_1 x'(t-t_0) + \dots + b_M x^{(M)}(t-t_0).
 \end{aligned}
 \quad \left. \vphantom{\begin{aligned} a_0 y(t-t_0) + a_1 y'(t-t_0) + \dots + a_N y^{(N)}(t-t_0) \\ = b_0 x(t-t_0) + b_1 x'(t-t_0) + \dots + b_M x^{(M)}(t-t_0). \end{aligned}} \right\} \text{--- (1)}$$



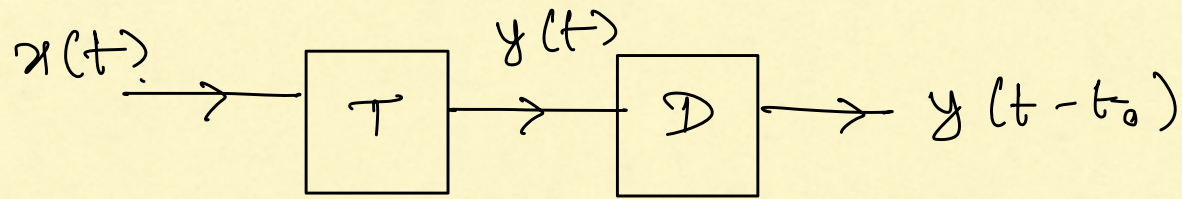
$$\begin{aligned}
 & a_0 z(t) + a_1 z'(t) + \dots + a_N z^{(N)}(t) \\
 &= b_0 x(t-t_0) + b_1 x'(t-t_0) + \dots + b_M x^{(M)}(t-t_0)
 \end{aligned}
 \quad \text{--- (2)}$$

Here, $w(t) = x(t-t_0)$

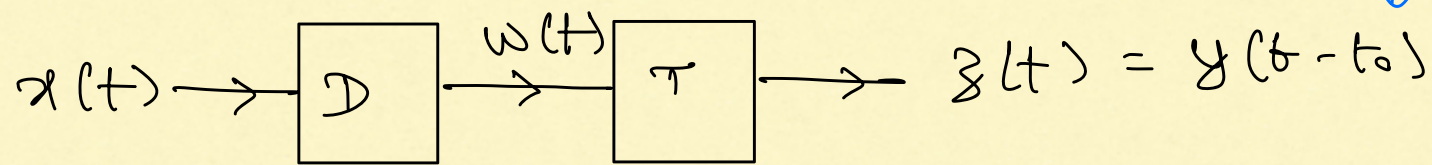
$$\Rightarrow w^{(k)}(t) = x^{(k)}(t-t_0)$$

Compare (1) & (2) :

$$z(t) = y(t-t_0)$$



Same output



$\therefore$ This system is time invariant.

Note:

The solution to the differential equation $\sum_{i=0}^N a_i y^{(i)}(t) = \sum_{j=0}^m b_j y^{(j)}(t)$ is unique only when additional information, known as auxiliary conditions, are specified.

In example 5b we have assumed that our system satisfies initial rest conditions, that is...

Initial Rest:

If for some $T \in \mathbb{R}$, we have

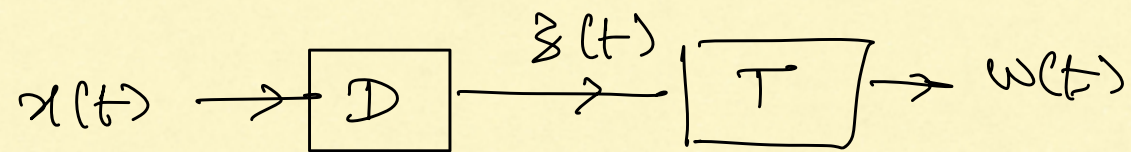
$$x(t) = 0 \quad \text{for all } t \leq T$$

then $y(t) = y'(t) = \dots = y^{(n)}(t) = 0$ for all $t \leq T$.

- * For RLC circuits this means that no current is flowing through the inductors and there is no charge across the capacitors as long as the input $x(t) = 0$, i.e., till time T .
- * A system described by linear constant coefficient differential equations is time invariant (and linear) under initial rest conditions.

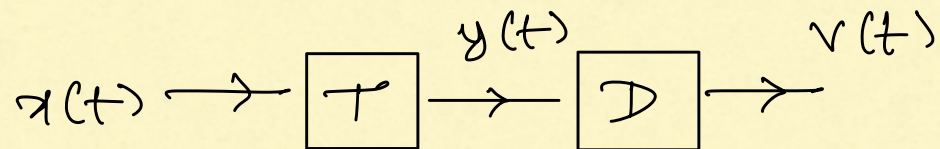
⑥ $x(t) \rightarrow \boxed{T} \rightarrow y(t) = \cos(\omega_0 t) \cdot x(t)$

Here, ω_0 is a constant.



$$z(t) =$$

$$w(t) =$$

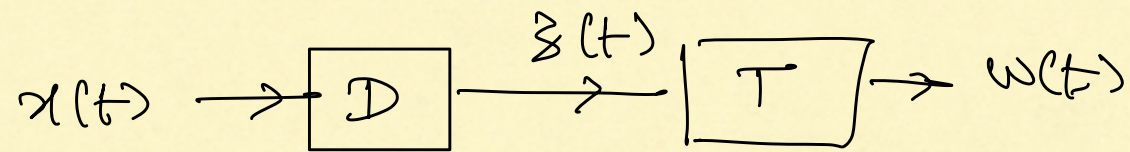


$$y(t) =$$

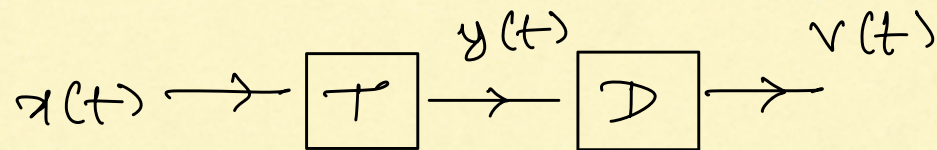
$$v(t) =$$

⑥ $x(t) \rightarrow \boxed{T} \rightarrow y(t) = \cos(\omega_0 t) \cdot x(t)$

Here, ω_0 is a constant.



$$\begin{aligned} w(t) &= z(t) \cdot \cos(\omega_0 t) \\ &= x(t - t_0) \cdot \cos(\omega_0 t) \end{aligned}$$



$$\begin{aligned} y(t) &= \cos(\omega_0 t) \cdot x(t) \\ v(t) &= y(t - t_0) \\ &= \cos(\omega_0 t - \omega_0 t_0) \\ &\quad \cdot x(t - t_0) \end{aligned}$$

$\therefore$ This system is NOT time invariant.

Other Examples

⑦ $x(t) \rightarrow \boxed{T} \rightarrow x(-t)$

Time-Invariant?

No.

⑧ $x(t) \rightarrow \boxed{T} \rightarrow x(at)$

No

IP. LTI Systems

Defn: A system is called linear time invariant or linear translation invariant or linear shift invariant if the system is both linear and time invariant.

To check for LTI property:

- ① Check for linearity
- ② Check for time invariance

} System must satisfy both.

If T is an LTI system

Suppose $x_1 \rightarrow \boxed{T} \rightarrow y_1$ and $x_2 \rightarrow \boxed{T} \rightarrow y_2$

$x_1(t-t_1) \rightarrow \boxed{T} \rightarrow$

$x_2(t-t_2) \rightarrow \boxed{T} \rightarrow$

$\alpha_1 x_1(t-t_1) + \alpha_2 x_2(t-t_2) \rightarrow \boxed{T} \rightarrow$

If T is an LTI system

Suppose $x_1 \rightarrow \boxed{T} \rightarrow y_1$ and $x_2 \rightarrow \boxed{T} \rightarrow y_2$

$x_1(t-t_1) \rightarrow \boxed{T} \rightarrow y_1(t-t_1)$

$x_2(t-t_2) \rightarrow \boxed{T} \rightarrow y_2(t-t_2)$

$\alpha_1 x_1(t-t_1) + \alpha_2 x_2(t-t_2) \rightarrow \boxed{T} \rightarrow \alpha_1 y_1(t-t_1) + \alpha_2 y_2(t-t_2)$

① Delay in inputs
 $\Rightarrow$ corrsg. delay in outputs.

② Linear combination of inputs
 $\Rightarrow$ corrsg. linear combination of outputs.

Examples of LTI Systems

$$\textcircled{1} \quad y(t) = a_1 x(t-t_1) + a_2 x(t-t_2) + \dots + a_k x(t-t_k)$$

$$\textcircled{2} \quad a_0 y(t) + \dots + a_N y^{(N)}(t) = b_0 x(t) + \dots + b_m x^{(m)}(t)$$

assuming initial rest conditions.

$$\textcircled{3} \quad y = x * h \quad \text{i.e.} \quad y(t) = \int_{-\infty}^{+\infty} x(\tau) \cdot h(t-\tau) d\tau$$

where h is a known signal.

All these systems are LTI.

Relation between LTI systems and Convolution

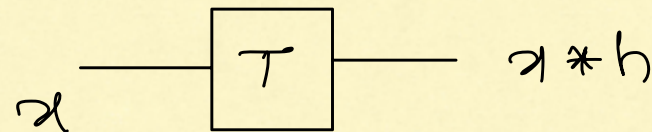
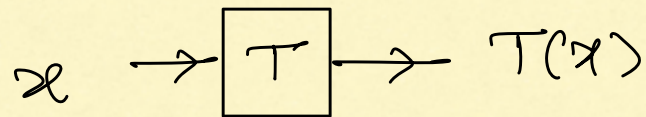
Suppose T is an LTI system and x is its input.

Theorem: (Under some mathematical conditions on T and x)

We can associate a unique signal h to the system T

Such that:

$$T(x) = x * h.$$



Ref: Stein & Weiss "Introduction to Fourier Analysis on Euclidean Spaces."

Chapter 1, Theorem 3.16.